Virginia Soda Tax Revenue Estimate

Proposed 5.0% Statewide Soda Tax Beginning January 2025

Illustrative Summary Report

Scenario date: January 1, 2025

Prepared by: Wataru Hoeltermann

1. Bottom-Line Estimates

Measure	Estimate	Interpretation
Estimated 2025 soda tax revenue	**$83,996,983**	Projected 2025 full-year revenue from a 5% Virginia soda tax, after accounting for the estimated demand response.
No-tax baseline soda consumption	2,328,279,060	2024 scaled soda units sold, used as the baseline for 2025 tax scenario
Estimated soda consumption after tax response	2,229,519,348	Estimated units sold after applying demand response to the baseline
Estimated reduction in consumption	**98,759,712**	Estimated decline in annual soda units sold relative to the no-tax baseline.

2. Price Elasticity of Soda Demand by Income Tier

A constant elasticity log-log specification was used:

ln(soda units sold) = a + b ln(soda price per unit)

Where the coefficient b is interpreted as the price elasticity of demand.

Income Tier	Elasticity
High (100k and above)	-0.71
Medium (40k to 100k)	-0.93
Low (less than 40k)	-1.94

Full-sample elasticity estimates were negative across all income tiers, with Low-income ZIP/ZCTA stores showing the strongest price response. In the state-level diagnostic results, **High-income stores** in Virginia and West Virginia showed **positive elasticities** of 0.51 and 0.69, respectively. Because positive elasticities are inconsistent with the expected downward-sloping demand relationship and appeared only in narrower state-tier cuts, these results were treated as anomalies and flagged for caution rather than used in the main revenue estimate.

3. Key Findings

Finding 1	Under stated assumptions, the proposed 5% Virginia soda tax is estimated to generate approximately **$84.0 million** in **2025 revenue.**
Finding 2	Estimated 2025 soda consumption falls by about **98.8 million units**, or roughly by **4.24%**, relative to the no-tax baseline.
Finding 3	**Medium-income** ZIP stores drive the largest share of revenue and unit reduction because they account for most Virginia soda sales in the sample
Finding 4	Estimated price elasticities are negative across income tiers, supporting the use of separate High, Medium, and Low income-tier demand responses.

4. Key Assumptions

Assumption	Description
1) Full tax pass-through	Assumes 100% of the tax is reflected in consumer soda prices, increasing consumer soda prices by 5%.
2) Representative scaling	The 33 Virginia sample stores are treated as representative of 1,000 statewide stores.
3) Income-tier price elasticities	High, Medium, and Low ZIP/ZCTA price elasticities are applied separately, then summed for the statewide estimate
4) Price-driven demand response	Revenue is calculated from soda sales after the estimated unit reduction as an expected response to a price increase.
5) Unit of Analysis	analysis unit provided in the assignment, not as fluid ounces, servings, or containers of a known size. Therefore, the consumption estimate is reported as a change in

5. Method in Brief

Step	Description
1. Complete data inputs	Pulled 2024 ACS 5-year median household income data from the Census API using table "B19013" at ZIP-level. Combined 2022–2024 income data to calculate three-year average income and classify each ZIP/ZCTA into the appropriate income tier.
2. Build analytical dataset	Used Power Query to reshape the supplied wide monthly sales sheets into a store-month fact table, then linked each record to store, date, state, ZIP/ZCTA income, and income-tier fields for pivoting and slicing.
3. Explore patterns and anomalies	Reviewed sales, soda units, soda prices, geography, income tiers, and time patterns using PivotTables and charts. I expected higher prices to reduce quantity demanded, but also checked for unstable estimates caused by small store counts or state-tier anomalies.
4. Estimate demand response	Estimated full-sample soda price elasticity by income tier using a log-log demand specification relating soda units sold to soda price per unit. These elasticity estimates were used as the behavioral response inputs for the tax scenario.
5. Estimate Virginia tax impact	Used 2024 Virginia soda units and sales as the no-tax baseline, scaled the 33 Virginia sample stores to the assignment assumption of 1,000 stores, applied the proposed 5.0% tax under full pass-through, and calculated post-response tax revenue and the change in soda consumption.

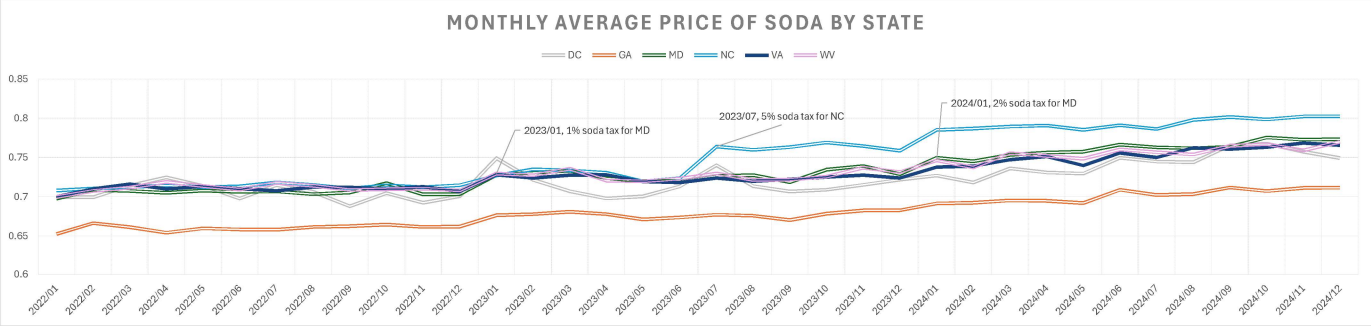
Virginia Soda Tax Revenue Estimate

Proposed 5.0% Statewide Soda Tax Beginning January 2025

Prepared by: Wataru Hoeltermann

Illustrative Summary Report

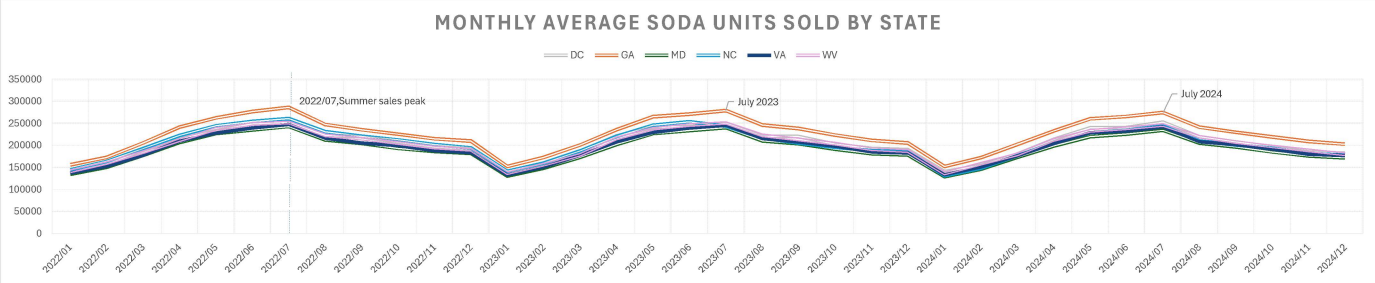
Scenario date: January 1, 2025



Graphs are illustrated from sample set given.

Takeaways:

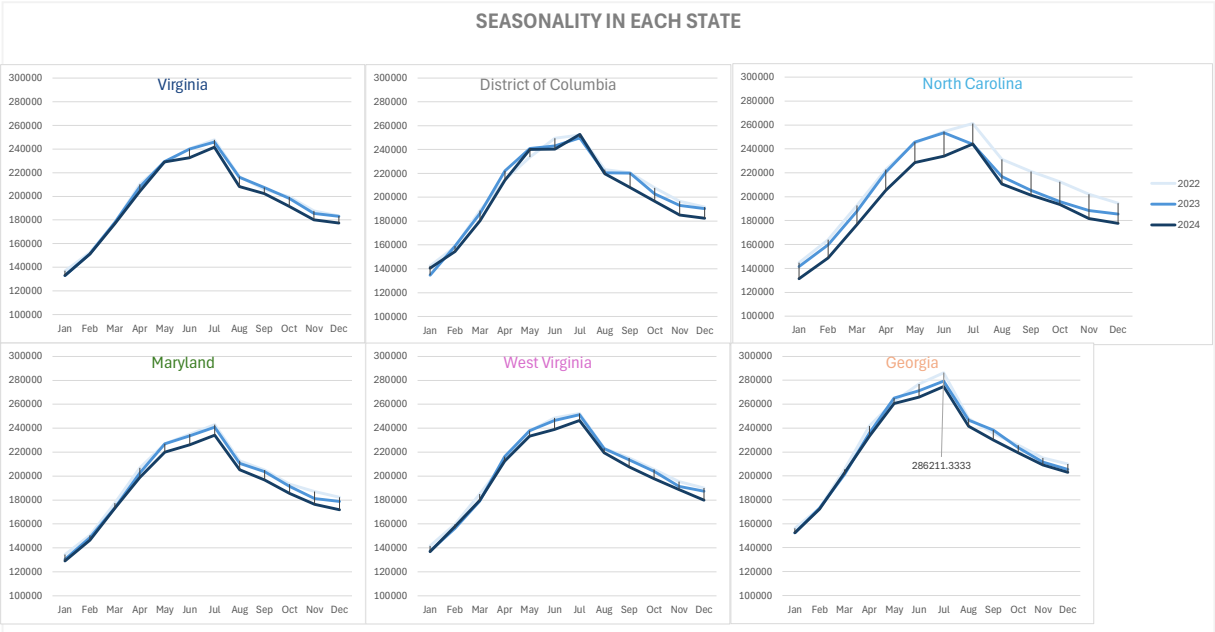
Price per soda is mostly stable, with notable tax-period change to NC due to a large 5% soda tax. MD and DC both have a much less demonstrated change, albeit a lower soda tax, they end around the same price per unit as non-taxed states such as VA and WV.



Takeaways:

Soda units sold show clear seasonality, with a rise in sales during summer and dip during winter.

Each state follows a very similar timing, with Georgia consuming approximately 40k more than other states during summertime.
North Carolina's tax effect is very apparent, with a visible dip in consumption after the tax policy on July, 2023.



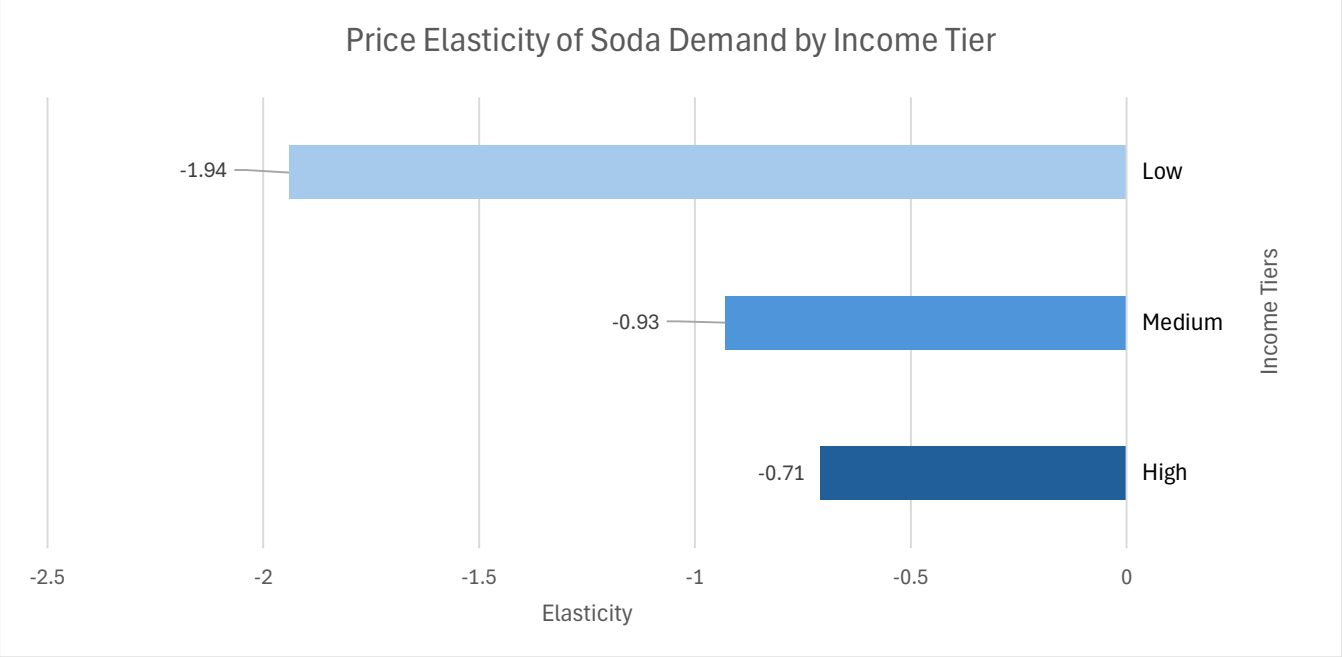
Virginia Soda Tax Revenue Estimate

Proposed 5.0% Statewide Soda Tax Beginning January 2025

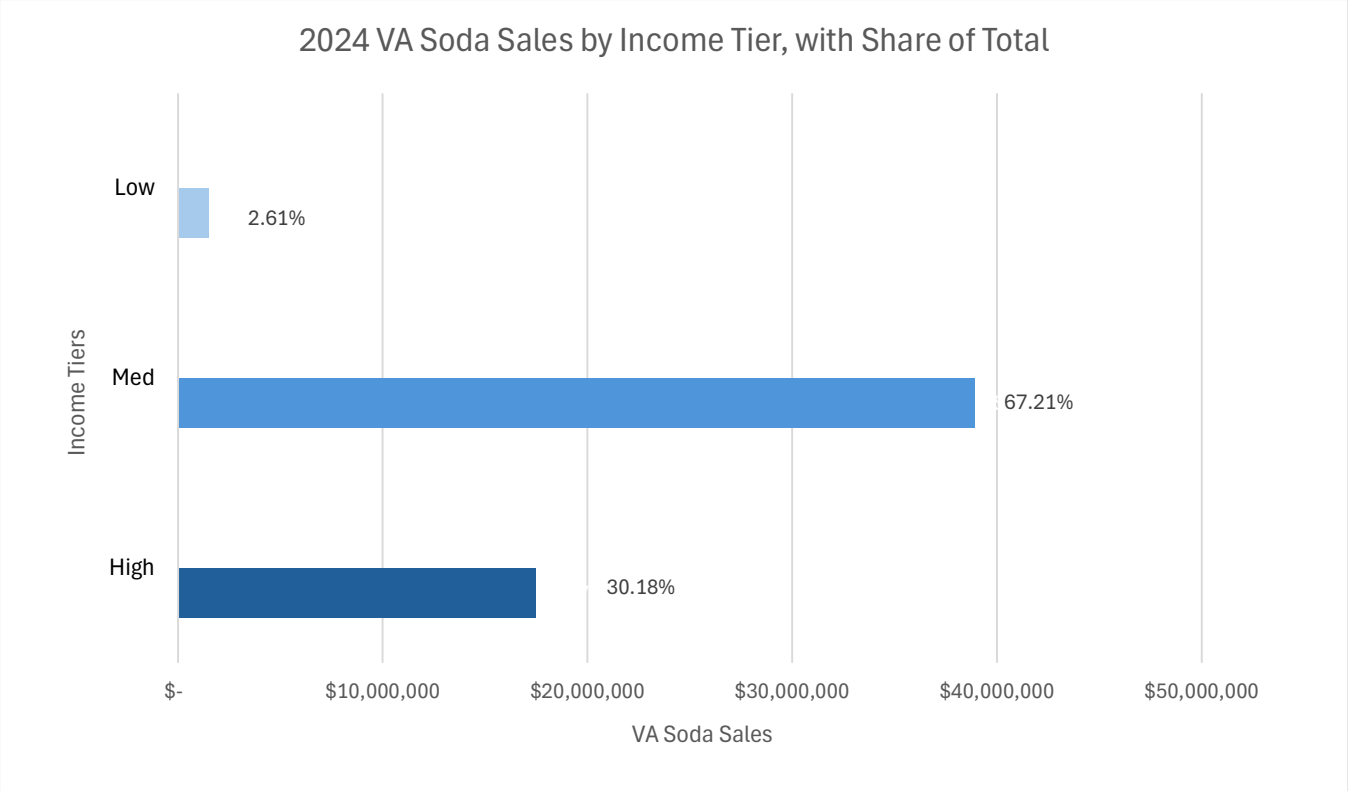
Prepared by: Wataru Hoeltermann

Illustrative Summary Report

Scenario date: January 1, 2025



Low-income ZIP/ZCTA stores show the strongest estimated price response, while High and Medium tiers are less responsive.



Stores in Low-income neighborhoods has the strongest estimated price response, while in High- and Medium-income neighborhoods are less responsive.

Virginia Soda Tax Revenue Estimate

Proposed 5.0% Statewide Soda Tax Beginning January 2025

Illustrative Summary Report

Prepared by: Wataru Hoeltermann

Scenario date: January 1, 2025

Main VA Estimate: 100% Pass-Through Scenario

Income Tier	VA Sample Store Count	Soda Price Per Unit	Scaled (33 to 1000 stores) 2024 Soda Sales	Elasticity	Estimated Units After Demand Shift	Estimated Unit Change	Estimated Tax Revenue	Notes
High	11	$ 0.75	$529.5M	-0.71	676.9M	-23.9M	$25.6M	Using VA-only High income tier elasticity produced an elastic (0.51) estimate, but this was only flagged as anomalous and not used for estimations.
Medium	21	$ 0.75	$1,179.1M	-0.93	1,497.9M	-69.4M	$56.3M	
Low	1	$ 0.76	$45.8M	-1.94	54.8M	-5.4M	$2.1M	Only one VA Low income store was observed, so the upscaled estimate should be treated with caution.
Total	**33**	**$ 0.75**	**$1,754.4M**	**Tier-summed**	**2,229.5M**	**-98.8M**	**$84.0M**	

Notes: Estimated demand response is based on elasticity and the assumed price increase. Tier-level estimates should be treated with caution as sample sizes differ, with the total row being our preferred estimates for revenue and demand effects.

Pass-through sensitivity summary

Pass-through	Consumer Price Increase	Post-Tax Units	Unit Change	Est. Tax Revenue
100%	5.00%	2,229.5M	(98.8M)	**$84.0M**
75%	3.75%	2,253.3M	(74.9M)	**$84.9M**
50%	2.50%	2,277.7M	(50.5M)	**$85.8M**

Notes: Consistent trend of lower pass-through produces higher estimated tax revenue, as the smaller consumer price increase leads to a smaller reduction in quantity demanded and thus a larger taxable sales base.

Sensitivity results are illustrative and depend on the assumed pass-through rates.

Virginia Soda Tax Revenue Estimate

Proposed 5.0% Statewide Soda Tax Beginning January 2025

Illustrative Summary Report

Prepared by: Wataru Hoeltermann

Scenario date: January 1, 2025

6. Interpretations and Limitations

These estimates should be interpreted as an illustrative fiscal scenario, not as a full causal evaluation of soda tax policy. Under the stated assumptions, the proposed 5% Virginia soda tax is projected to raise revenue while reducing estimated soda units sold. The taxable base

Estimate results could differ under alternative scaling methods, different pass-through rates, or store-weighting methods. ZIP/ZCTA-level income tiers represent store information and may not necessarily be capturing household-level tax burden in that area, as people may shop in different zip codes than their home. Elasticity estimates are also based on observed price and quantity patterns, not accounting for random price variation, so it may not reflect product mix, seasonality, store differences, or other omitted factors. This model also does not account for substitution, cross-border shopping, health outcomes, or long-run behaviors.

7. Further Analysis

Further Analysis	Purpose
Model specification tests	Estimate alternative models with store fixed effects, month/year controls, and income-tier interactions to test whether elasticity estimates are stable.
Geographic detail	Map ZIP/ZCTA-level patterns and test whether results vary by border areas, urban/rural location, or nearby taxed jurisdictions.
Product and unit definition	Clarify if the unit of analysis represents bottles, cases, ounces, or another measurement. We can test whether excise (volume-based) tax estimates differ from price-based.
Substitution and behavior	Examine substitution with other beverages, cross-state shopping, seasonality impacts, long-run behavioral adjustments, and store-level heterogeneity.
Full fiscal impact model	Extend the scenario to include administrative costs, compliance costs, revenue usage, household tax burden, and broader distributional effects.